

Introduction

1.1 High-Dimensional Data

The sustained development of technologies, data storage resources, and computing resources give rise to the production, storage, and processing of an exponentially growing volume of data. Data are ubiquitous and have a dramatic impact on almost every branch of human activities, including science, medicine, business, finance and administration. For example, wide-scale data enable to better understand the regulation mechanisms of living organisms, to create new therapies, to monitor climate and biodiversity changes, to optimize the resources in the industry and in administrations, to personalize the marketing for each individual consumer, etc.

A major characteristic of modern data is that they often record simultaneously thousands up to millions of *features* on each *object* or *individual*. Such data are said to be *high-dimensional*. Let us illustrate this characteristic with a few examples. These examples are relevant at the time of writing and may become outdated in a few years, yet we emphasize that the mathematical ideas conveyed in this book are independent of these examples and will remain relevant.

- **Biotech data:** Recent biotechnologies enable to acquire high-dimensional data on single individuals. For example, DNA microarrays measure the transcription level¹ of tens of thousands of genes simultaneously; see Figure 1.1. Next generation sequencing devices (NGS) improve on these microarrays by allowing to sense the “transcription level” of virtually any part of the genome. Similarly, in proteomics some technologies can gauge the abundance of thousands of proteins simultaneously. These data are crucial for investigating biological regulation mechanisms and creating new drugs. In such biotech data, the number p of “variables” that are sensed scales in thousands and is most of the time much larger than the number n of “individuals” involved in the experiment (number of repetitions, rarely exceeding a few hundreds).
- **Images (and videos):** Large databases of images are continuously collected all around the world. They include medical images, massive astrophysic images, video surveillance images, etc. Each image is made of thousands up to millions of

¹The transcription level of a gene in a cell at a given time corresponds to the quantity of ARNm associated to this gene present at this time in the cell.

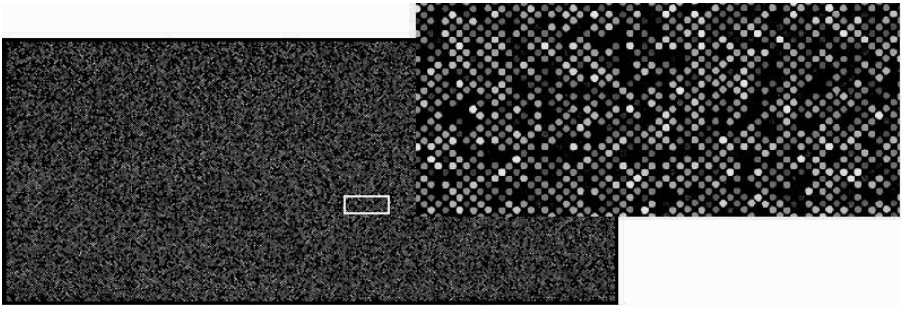


Figure 1.1 Whole human genome microarray covering more than 41,000 human genes and transcripts on a standard 1" $\times$ 3" glass slide format. © Agilent Technologies, Inc. 2004. Reproduced with permission, courtesy of Agilent Technologies, Inc.

pixels or voxels. For medical images, as for biotech data, the number p of pixels can be much larger than the number n of patients in the cohort under study.

- **Consumers preferences data:** Websites and loyalty programs collect huge amounts of information on the preferences and the behaviors of customers. These data are processed for marketing purposes, for recommendations, and for fixing personalized prices. For example, recommendation systems (for movies, books, music, etc.) gather the customers' ratings on various products, together with some personal data (age, sex, location) and guess from them which products could be of interest for a given consumer.
- **Business data:** Every major company has its own chief data officer who supervises the optimal exploitation of internal and external data. For example, logistic and transportation companies intensively process internal and geo-economic data in order to optimize the allocation of their resources and to try to forecast precisely the future demand. Insurance companies widely rely on various sources of data in order to control their risk and allocate at best their financial resources. Many profitable activities of the financial industry are based on the intensive processing of transaction data from all over the world. Again, the dimensionality of the data processed in these examples can scale in thousands.
- **Crowdsourcing data:** The launch of websites dedicated to participative data recording together with the spreading of smartphones enable volunteers to record online massive data sets. For example, the Cornell Lab of Ornithology and the National Audubon Society have jointly launched a crowdsourcing program "eBird" <http://ebird.org> inviting all bird-watchers from North America to record via an online checklist all the birds they have seen and heard during their last birding session. The purpose of this program is to monitor birds abundances and their evolutions across North America. In 2014, eBird involved tens of thousands of participants, which had already recorded millions of observations.

Blessing?

Being able to sense simultaneously thousands of variables on each “individual” sounds like good news: Potentially we will be able to scan every variable that may influence the phenomenon under study. The statistical reality unfortunately clashes with this optimistic statement: **Separating the signal from the noise is in general almost impossible in high-dimensional data.** This phenomenon described in the next section is often called the “curse of dimensionality.”

1.2 Curse of Dimensionality

The impact of high dimensionality on statistics is multiple. First, high-dimensional spaces are **vast** and data points are **isolated** in their immensity. Second, the accumulation of small fluctuations in many different directions can **produce a large global fluctuation**. Third, an event that is an **accumulation of rare events** may not be rare. Finally, numerical **computations and optimizations** in high-dimensional spaces can be overly **intensive**.

1.2.1 Lost in the Immensity of High-Dimensional Spaces

Let us illustrate this issue with an example. We consider a situation where we want to explain a **response variable** $Y \in \mathbb{R}$ by p variables $X_1, \dots, X_p \in [0, 1]$. Assume, for example, that each variable X_k follows a uniform distribution on $[0, 1]$. If these variables are independent, then the variable $X = (X_1, \dots, X_p) \in [0, 1]^p$ follows a uniform distribution on the hypercube $[0, 1]^p$. Our data consist of n i.i.d. observations $(Y_i, X^{(i)})_{i=1, \dots, n}$ of the variables Y and X . We model them with the classical regression equation

$$Y_i = f(X^{(i)}) + \varepsilon_i, \quad i = 1, \dots, n,$$

with $f: [0, 1]^p \rightarrow \mathbb{R}$ and $\varepsilon_1, \dots, \varepsilon_n$ independent and centered.

Assuming that the function f is smooth, it is natural to estimate $f(x)$ by some average of the Y_i associated to the $X^{(i)}$ in the vicinity of x . The most simple version of this idea is the **k -Nearest Neighbors estimator**, where $f(x)$ is estimated by the mean of the Y_i associated to the k points $X^{(i)}$, which are the nearest from x . Some more sophisticated versions of this idea use a weighted average of the Y_i with weights that are **a decreasing function of the distance $\|X^{(i)} - x\|$ (like kernel smoothing)**. The basic idea being in all cases to use a **local average** of the data. This idea makes perfect sense in low-dimensional settings, as illustrated in Figure 1.2.

Unfortunately, when the dimension p increases, the notion of “nearest points” vanishes. This phenomenon is illustrated in Figure 1.3, where we have plotted the histograms of the distribution of the pairwise-distances $\{\|X^{(i)} - X^{(j)}\| : 1 \leq i < j \leq n\}$ for $n = 100$ and dimensions $p = 2, 10, 100$, and 1000. When the dimension p increases, we observe in Figure 1.3 that

- the minimal distance between two points increases,



Figure 1.2 *Canadian high school graduate earnings from the 1971 Canadian Census Public Use Tapes. Black dots: records. Gray line: 25-Nearest Neighbors estimator. Dashed line: local averaging by kernel smoothing.*

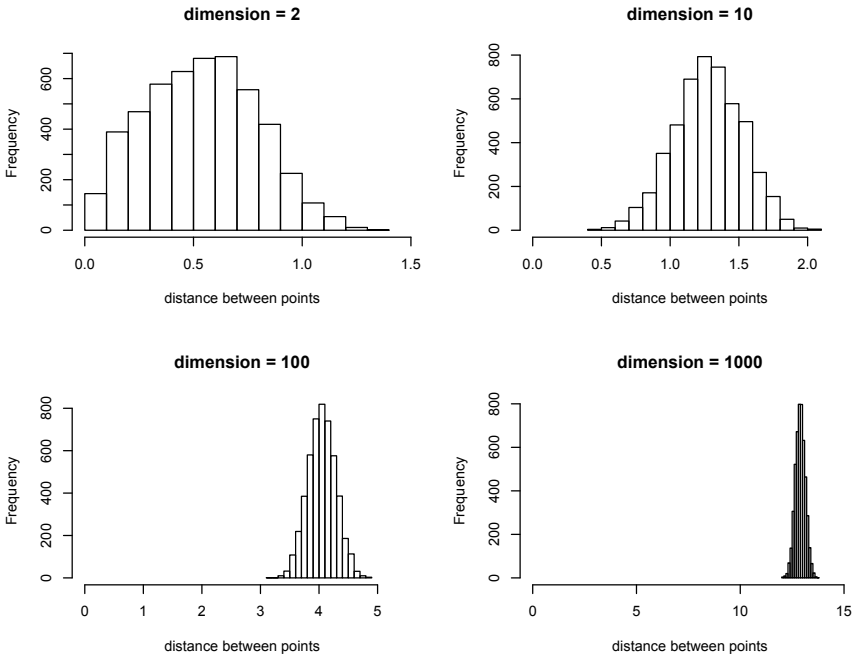


Figure 1.3 *Histograms of the pairwise-distances between $n = 100$ points sampled uniformly in the hypercube $[0, 1]^p$, for $p = 2, 10, 100$, and 1000 .*

- all the points are at a similar distance one from the others, so the notion of “nearest-points” vanishes.

In particular, any estimator based on a local averaging will fail with such data.

Let us quantify roughly the above observations. Writing U and U' for two independent random variables with uniform distribution on $[0, 1]$, the mean square distance between $X^{(i)}$ and $X^{(j)}$ is

$$\mathbb{E} [\|X^{(i)} - X^{(j)}\|^2] = \sum_{k=1}^p \mathbb{E} \left[\left(X_k^{(i)} - X_k^{(j)} \right)^2 \right] = p \mathbb{E} [(U - U')^2] = p/6,$$

and the standard deviation of this square distance is

$$\text{sdev} [\|X^{(i)} - X^{(j)}\|^2] = \sqrt{\sum_{k=1}^p \text{var} \left[\left(X_k^{(i)} - X_k^{(j)} \right)^2 \right]} = \sqrt{p \text{var} [(U' - U)^2]} \approx 0.2\sqrt{p}.$$

In particular, we observe that the typical square distance between two points sampled uniformly in $[0, 1]^p$ grows linearly with p , while the scaled deviation $\text{sdev} [\|X^{(i)} - X^{(j)}\|^2] / \mathbb{E} [\|X^{(i)} - X^{(j)}\|^2]$ shrinks like $p^{-1/2}$.

How Many Observations Do We Need?

Figure 1.3 shows that if the number n of observations remains fixed while the dimension p of the observations increases, the observations $X^{(1)}, \dots, X^{(n)}$ get rapidly very isolated and local methods cannot work. If for any $x \in [0, 1]^p$ we want to have at least one observation $X^{(i)}$ at distance less than one from x , then we must increase the number n of observations. How should this number n increase with the dimension p ? We investigate below this issue by computing a lower bound on the number n of points needed in order to fill the hypercube $[0, 1]^p$ in such a way that at any $x \in [0, 1]^p$ there exists at least one point at distance less than 1 from x .

The volume $V_p(r)$ of a p -dimensional ball of radius $r > 0$ is equal to (see Exercise 1.6.2)

$$V_p(r) = \frac{\pi^{p/2}}{\Gamma(p/2 + 1)} r^p \underset{p \rightarrow \infty}{\sim} \left(\frac{2\pi e r^2}{p} \right)^{p/2} (p\pi)^{-1/2}, \quad (1.1)$$

where Γ represents the Gamma function $\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$ for $x > 0$.

If $x^{(1)}, \dots, x^{(n)}$ are such that for any $x \in [0, 1]^p$ there exists a point $x^{(i)}$ fulfilling $\|x^{(i)} - x\| \leq 1$, then the hypercube is covered by the family of unit balls centered in $x^{(1)}, \dots, x^{(n)}$,

$$[0, 1]^p \subset \bigcup_{i=1}^n B_p(x^{(i)}, 1).$$

As a consequence, the volume of the union of the n unit balls is larger than the volume of the hypercube, so $1 \leq nV_p(1)$. According to Equation (1.1), we then need at least

$$n \geq \frac{\Gamma(p/2 + 1)}{\pi^{p/2}} \underset{p \rightarrow \infty}{\sim} \left(\frac{p}{2\pi e} \right)^{p/2} \sqrt{p\pi} \quad (1.2)$$

The Strange Geometry of High-Dimensional Spaces (I)

High-dimensional balls have a vanishing volume!

From the Formula (1.1), we observe that for any $r > 0$, the volume $V_p(r)$ of a ball of radius r goes to zero more than exponentially fast with the dimension p . We illustrate this phenomenon in Figure 1.4 by plotting $p \rightarrow V_p(1)$. We observe that for $p = 20$ the volume of the unit ball is already almost 0.

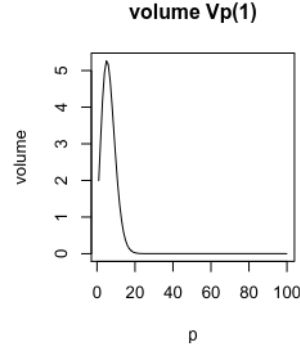


Figure 1.4

points in order to fill the hypercube $[0, 1]^p$. This number of points then grows more than exponentially fast with p . If we come back to our above example in the regression setting, it means that if we want a local average estimator to work with observations uniformly distributed in $[0, 1]^p$ with p larger than a few tens, then we would need a number n of observations, which is completely unrealistic (see Table 1.1).

p	20	30	50	100	150	200
n	39	45630	$5.7 \cdot 10^{12}$	$42 \cdot 10^{39}$	$1.28 \cdot 10^{72}$	larger than the estimated number of particles in the observable universe

Table 1.1 Lower Bound (1.2) on the required number of points for filling the hypercube $[0, 1]^p$.

The moral of this example is that we have to be very careful with our geometric intuitions in high-dimensional spaces. These spaces have some counterintuitive geometric properties, as illustrated in Figures 1.4 and 1.5.

1.2.2 Fluctuations Cumulate

Assume that you want to evaluate some function $F(\theta_1)$ of some scalar value $\theta_1 \in \mathbb{R}$. Assume that you have only access to a noisy observation of θ_1 , denoted by $X_1 = \theta_1 + \varepsilon_1$, with $\mathbb{E}[\varepsilon_1] = 0$ and $\text{var}(\varepsilon_1) = \sigma^2$. If the function F is 1-Lipschitz, then the mean square error is

$$\mathbb{E}[\|F(X_1) - F(\theta_1)\|^2] \leq \mathbb{E}[|\varepsilon_1|^2] = \sigma^2.$$

In particular, if the variance σ^2 of the noise is small, then this error is small.

Assume now that you need to evaluate a function $F(\theta_1, \dots, \theta_p)$ from noisy observations $X_j = \theta_j + \varepsilon_j$ of the θ_j . Assume that the noise variables $\varepsilon_1, \dots, \varepsilon_p$ are all centered